MICHELLE J. WANG

Student | Aspiring Engineer

MAKER | 4-H'ER | TEAM PLAYER | ADVOCATE

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Michelle Wang

TreeKangaroo

EDUCATION

08/19 – 05/23 **Carbondale Community High School (CCHS)** **Class of 2023**

- GPA: **4.0/4.0** (unweighted), **4.44/4.44** (weighted) | Class Rank: **1**
- SAT: **1580** - 780 ERW, 800 Math (April 2022)
- ACT: **36** - all subscores 36 (June 2022)
- AP Courses **(scored 5 on all exams)**: Physics: Mechanics (11th), Physics: E&M (11th), Calculus BC (11th), Biology (11th), United States History (11th), Physics 2 (10th), World History (10th), Physics 1 (9th), Chinese Language & Culture (9th)
- AIME Qualifier (2022)

MAKER/STEM EXPERIENCE

09/21 – present **Autonomous Drone** **Independent Research**

Project repository

- Developed an X500 frame quadcopter with Jetson Xavier GPU onboard. The developed software performs image capture & pre-processing, object detection (using YOLOv4), object pose estimation (using a neural network I developed), and navigation control. Successfully tracks objects of interest in flight
- Won **2nd place oral presentation at National Junior Science and Humanities Symposium** (JSHS) in Engineering & Technology and **1st place** at IL JSHS overall

09/19 – 05/21 **IMU-based Wearable Device** **Independent Research**

Project repository

- Developed an Arduino and IMU-based wearable device and sensor data processing algorithms for tracking knee angle during walking. Won **3rd place** at IL JSHS overall and national qualifier (2020)
- Developed a convolutional neural network to classify running form based on data collected by the wearable. Won **1st place** at IL JSHS overall and national qualifier (2021)

11/20 – 02/21 **AI & Sustainability Special Interest Group** **Changemakers in AI**

Press release

- Led (with a stipend) an 8-person team of AI4ALL alumni to use AI techniques to analyze publicly available agricultural data for predicting the sustainability of different counties

LEADERSHIP & COMMUNITY ENGAGEMENT

04/21 – present **Illinois 4-H Youth Leadership Team**

Team website

- 20 team members selected statewide to represent the 200,000 4-H members in Illinois. The team plans and implements statewide events, advises state 4-H activities, and lobbies legislators at the state capitol on behalf of 4-H
- I initiated the STEM Committee and serve as chair (2022)

01/20 – present **Illinois 4-H in Space Mission Command Team**
Press release

- Founding member selected from high school applicants from around the state
- Work with the Lab for Advanced Space Systems at Illinois (LASSI) to develop, test, and eventually teach a CubeSat centered curriculum

07/19 – present **4-H Jackson County STEM Club Youth Leader**
Club homepage

- Led a series of hands-on CircuitPython workshops for 4-10th graders (9/2022). **Secured $500 (NCWIT AspireIT Impact award)** to purchase hardware for all participants
- Led a series of hands-on drone and block coding workshops for middle schoolers in 2021 (August-Sept.) and 2022 (May-June). **Secured $1987 (JSHS Momentum grant)** to purchase 10 Tello EDU drones for the workshops
- Led virtual 5-county STEM Club for 3-10th graders (10/2020 – 4/2021). **Secured $500 (AI4Impact grant)** to purchase hands-on project supplies for all participants
- Taught Python programming to 25 Jackson County 4-H Computing Club members (6/2019-8/2019)

08/21 – present **Co-founder/Leader of CCHS Maker Club**

- Plan monthly general meetings for 40 active members, train new leadership, and organize technical workshops
- Lead a 3D-printed prosthetic hand group and work with the e-NABLE project. **Secured $300 worth of hardware in donations** from 3D Universe for the project

09/19 – present **Lead Instructor of Carbondale Middle School (CMS) MathCounts Team**

- Coordinate scheduling with CMS teacher and organize a team of 8 high school volunteer instructors to teach the CMS math team weekly after school
- Plan competition math curriculum. In 2021, the CMS Math Team won 1st place at regionals for the first time in over 5 years.

08/20 – present **CCHS FBLA (Future Bussiness Leaders of America) Officer**

- Elected to and serve as **Treasurer (10th), Secretary (11th), and Vice President (12th)**
- Collaborate with other officers to plan meetings and events for 50-person chapter. Network with local businesses to secure support for FBLA activities

04/21 – 06/21 **Carbondale Boys and Girls Club STEM Program Instructor**
Project instructions

- Designed hands-on projects and taught a series of 6 STEM workshops to 4-5th graders

HONORS & AWARDS

04/22 **2nd place oral presentation at National JSHS in Engineering and Technology**: "An Autonomous Drone with Object Detection and Tracking Capabilities"

08/22 **NCWIT (National Center for Women and Information Technology) AspireIT Impact Award winner**: 50 winners chosen nationwide for significant efforts in educating underrepresented students in computing & technology

01/22 **NCWIT Aspirations in Computing National Honorable Mention**: 360 award recipients were selected out of more than 3700 applicants based on aptitude/aspirations in computing, demonstrated by computing experience, leadership, tenacity in the face of barriers to access, and future plans in computing

03/22 **1st place oral presentation overall at Illinois JSHS** and national JSHS qualifier and participant: "An Autonomous Drone with Object Detection and Tracking Capabilities"

03/21 **1st place oral presentation overall at Illinois JSHS** and national JSHS qualifier and participant: "Running Form Detection Using Artificial Neural Networks and Inertial Measurement Unit"

03/20 **3rd place oral presentation overall at Illinois JSHS** and national JSHS qualifier and participant: "An Integration-Free Accelerometer Based Approach for Measuring Peak Knee Angle"

05/22 **National History Day qualifier and participant**: Group performance

04/22 **Illinois State Academic Challenge Physics 2nd place**

04/21 **Illinois State Academic Challenge Physics 1st place**

05/20 **Illinois State FBLA Conference 2nd place** in E-Business category and national FBLA conference qualifier and participant

01/22 **Recipient of the Illinois 4-H Leadership & Civic Engagement Scholarship**: 5 recipients selected statewide

01/21 **Recipient of the Illinois 4-H STEM & Robotics Scholarship**: 5 recipients selected statewide

09/21 **3rd place and national qualifier at Illinois 4-H State Archery Shoot**: Recurve archery

SUMMER PROGRAMS

06/22 – 07/22 **Summer Science Program in Astrophysics** — **Summer Science Program**

Program website

- Conducted observations and developed Python programs to determine the orbit of a near-Earth asteroid in a team of 3
- Supported by SSP Scholarship and Garwin Family Foundation

07/2021 **MIT Beaver Works Summer Institute** — **MIT Lincoln Labs: Beaver Works**

Program website

- Developed Python programs to guide a small drone to autonomously navigate an obstacle racecourse in a team of 3. Won the **grand prize** in Autonomous Air Vehicle Racing program (project repository)

07/2020 **Stanford AI4ALL** — **Stanford Institute for Human-Centered Artificial Intelligence**

Program website

- Group research project using AI to analyze gene profile data for predicting cancer
- Attended on full scholarship from AI4ALL

SKILLS & INTERESTS

- **Programming:** Python, Arduino, HTML, C#
- **AI & Computer Vision Libraries:** Tensorflow, Keras, Scikit-learn, OpenCV
- **Software:** Robot Operating System (ROS), Unity, git, Unix, Autodesk Fusion 360, Sketchup, Adobe Photoshop & Illustrator, Lightworks, Audacity
- **Languages:** Chinese (fluent), Spanish (working proficiency)
- **Academic Interests:** Biomimicry, artificial intelligence, robotics
- **Other Interests:** Rock climbing, hiking, swimming, reading, wildlife, astronomy, cooking